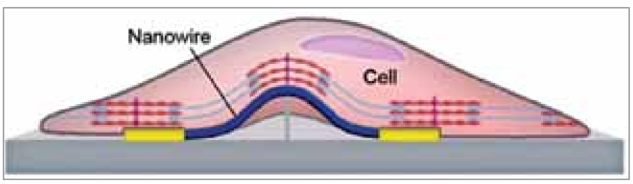
E-Tattoos that can measure blood pressure

Blood pressure is the most important vital sign to be continuously measured but methods outside the clinic, without cuffs, are very limited. Researchers at Texas A&M University and The University of Texas at Austin have developed an electronic tattoo that can be comfortably worn on the wrist for hours and provides continuous blood pressure measures with an accuracy level that surpasses practically all solutions on the market today. To obtain correct blood pressure measurements, the team had to develop a machine learning algorithm to examine the correlation between bioimpedance alterations and variations in blood volume as it was difficult to correlate without AI.

The continuous monitoring of the e-tattoo allows for blood pressure measurements in all kinds of situations (Credit: https://news.utexas.edu)

Nanolectronic sensor to monitor cardiac responses

The cell is a basic functional element in biology and analysing its mechanical and electrical behaviours help to understand cell state and, consequently, are important for health monitoring, disease diagnosis, and tissue repair. A research team from the University of Massachusetts has developed a mini sensor using a suspended nanowire that can measure electrical and mechanical cellular responses in cardiac tissue simultaneously. This sensor is assembled using a 3D suspended semiconducting silicon nanowire and can convert 'heard' bioelectrical and biomechanical activities into electrical sensing signals for detection. The idea of combining the multiple sensing functions in one device will also broaden the capabilities of general bio-interface engineering.



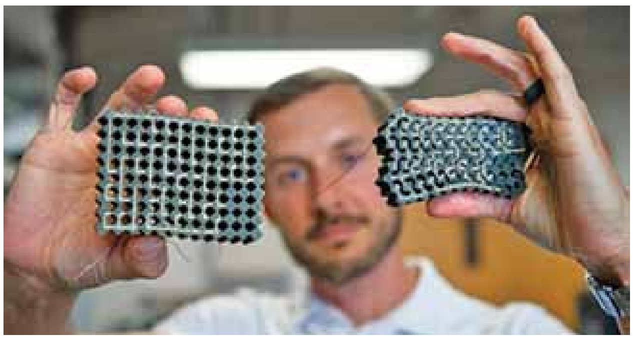
Schematic of cell-sensor interface (Credit: Jun Yao)

A handy extra arm for better grasping of objects

The team of researchers at Rice University's George R. Brown School of Engineering has designed and built robotic devices that are geared toward those living with disabilities. They have developed a handy extra limb for better grasping of objects. These arms are powered by compressed air. The prototype 'arm' is a piece of fabric that hugs the body and extends outward upon activation. The elastomer lining on it enables gripping the slippery objects. The team has also built a shirt with a bellows-like actuator that is attached to the armpit. This attachment enables the wearer to pick up a 4.5kg heavy object. This research can be taken forward by implementing the mathematical model to optimise performance for a specific user group.

A soft polymer IC material that can 'think' under stress

Researchers are always striving hard to obtain material that can work like the human brain. Earlier researched materials were limited to only logic gates operating on binary input-output signals and had no way to compute high-level logical operations that are central to integrated circuits. Researchers from Penn State have created an integrated circuit material that could 'think' under mechanical stress without additional circuits to process signals. They have discovered a soft polymer material that can act on its own. The soft, conductive polymer material has reconfigurable circuits that can realise combinational logic. The material uses a 'thinking' process like that of the human brain. It has potential applications in infrastructure repair, autonomous search-and-rescue systems, and in bio-hybrid materials.



Penn State researchers create mechanical integrated circuit materials from conductive and non-conductive rubber materials that sense and react to tactile input, such as force (Credit: Kelby Hochriether, Penn State)